



Essay / Assignment Title: Predictive Modelling for Credit Card Default Using Machine Learning

Programme title: MSc in Artificial Intelligence

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INTRODUCTION

Credit card default prediction is a critical application of Artificial Intelligence (AI) and Machine Learning (ML) within the banking and financial services industry. Financial institutions increasingly utilise predictive analytics to evaluate creditworthiness, minimise financial losses and support more informed lending decisions. Conventional credit assessment approaches typically rely on manual reviews and rule-based frameworks which may struggle to detect complex relationships and hidden patterns within large-scale datasets. In contrast, machine learning techniques can analyse customer demographics, financial records and repayment behaviour to identify individuals with a higher probability of defaulting on future credit obligations.

This project explores the application of machine learning methods for credit card default prediction. The dataset used in the study comprises 34788 customer records and approximately 30 variables including demographic information, billing details, repayment histories and engineered risk-related features. The primary objectives are to conduct Exploratory Data Analysis (EDA), preprocess and prepare the data for modelling, develop and compare multiple machine learning classification models, optimize model performance through hyperparameter tuning and interpret model outputs using feature importance techniques.

The growing adoption of machine learning in financial risk assessment is driven by its capability to process extensive structured datasets and identify complex nonlinear relationships that may not be captured by traditional statistical methods (James et al., 2021). Accordingly, this study evaluates five widely used classification algorithms to determine the most effective model for predicting credit card default risk and supporting credit decision-making processes.

CHAPTER ONE (the number of Chapters could be more depending on the content)

Research Methodology

A supervised machine learning methodology was employed to develop models capable of predicting credit card default risk. The overall process comprised six key phases as in data acquisition, exploratory data analysis (EDA), data preprocessing, model development, performance evaluation and model interpretation.

Initially, the dataset was analysed to investigate missing values, class distribution, outliers and potential relationships between variables. Data preparation involved handling missing observations through median imputation, standardising numerical features using the StandardScaler technique and removing non-predictive attributes such as the customer identification variable.

Subsequently, five classification algorithms were implemented and compared. They are - Logistic Regression, Support Vector Machine (SVM), Decision Tree, Random Forest and K-Nearest Neighbours (KNN). Model performance was evaluated using several metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. To further enhance predictive performance, hyperparameter tuning was applied while feature importance analysis was conducted to determine the variables that contributed most significantly to credit default prediction.

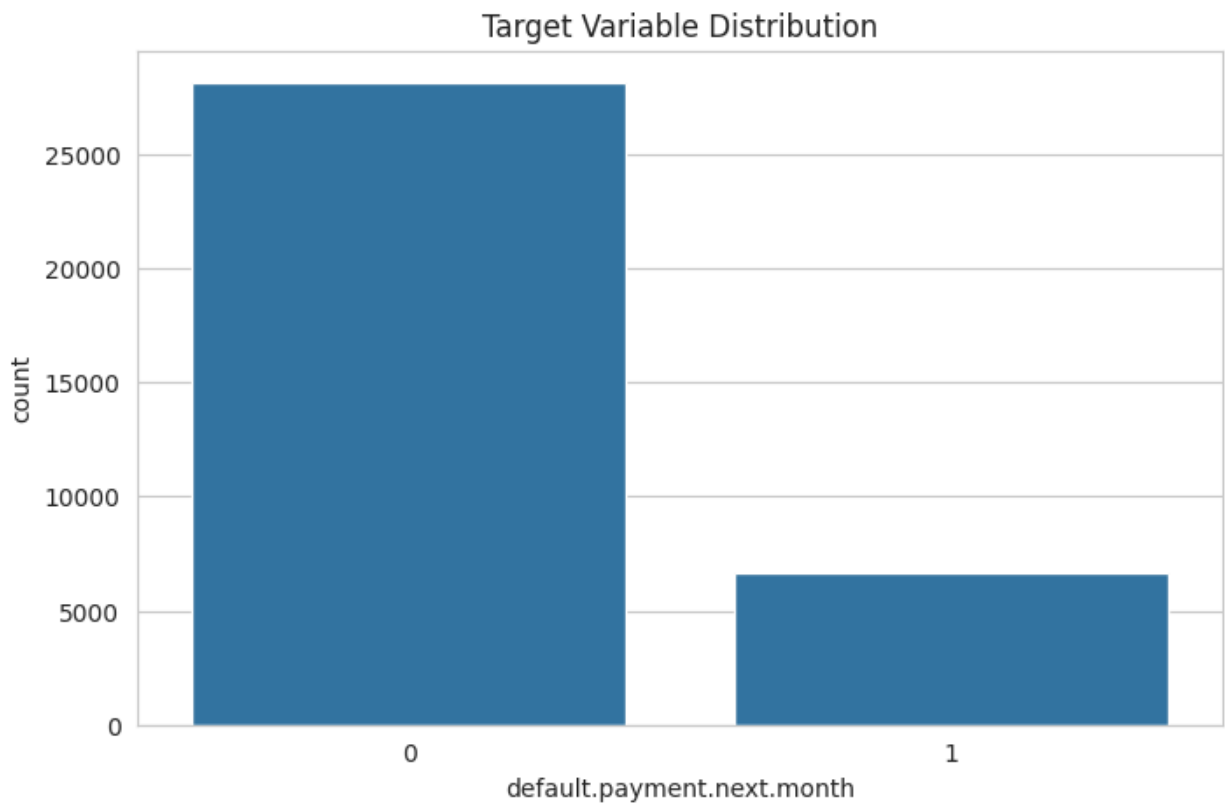
EXPLORATORY DATA ANALYSIS & DATA PREPARATION

1.1 Dataset Description

The dataset employed in this study consists of 34,788 customer records and approximately 30 variables. The dependent variable(**default.payment.next.month**) indicates whether a customer failed to meet their credit card repayment obligations in the subsequent month. Independent variables encompass demographic attributes, repayment history indicators, billing information, credit limits and several engineered features designed to capture risk-related characteristics.

1.2 Analysis of the Target Variable

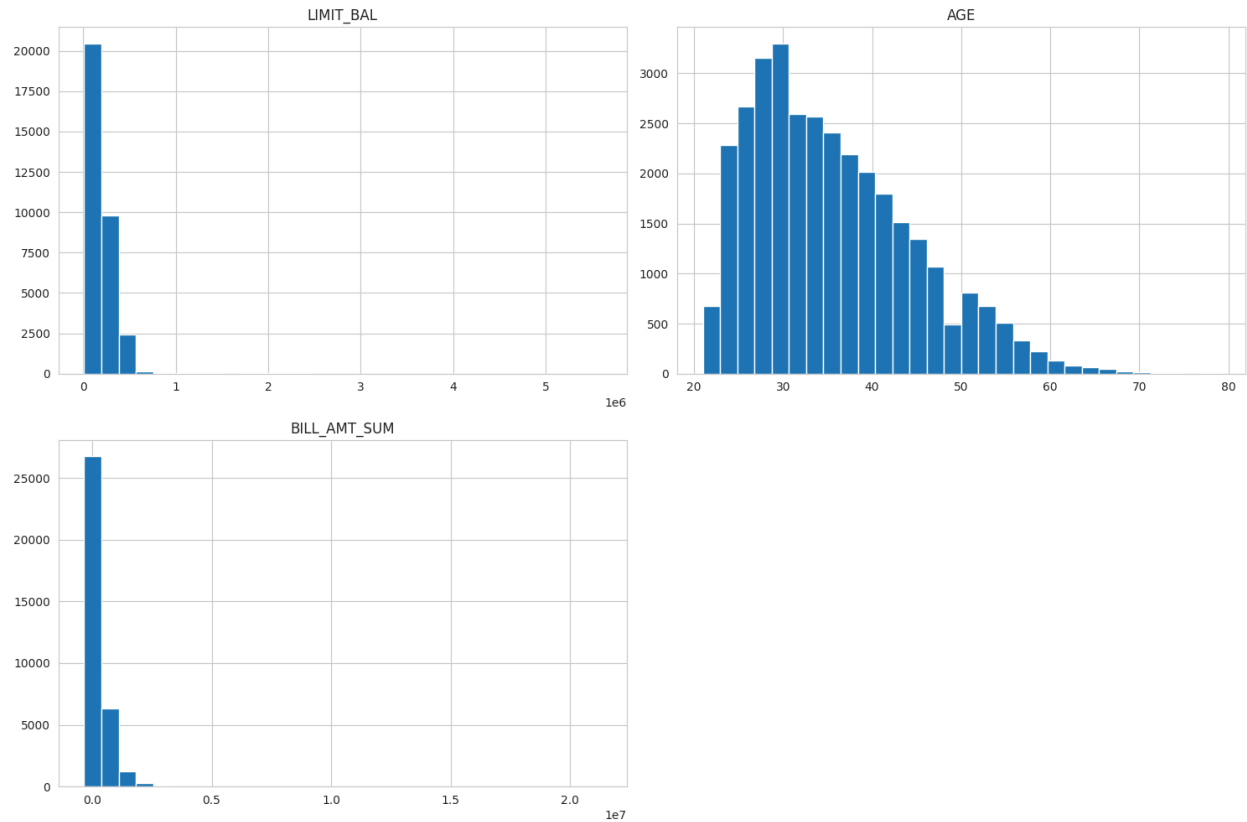
The initial phase of the exploratory data analysis (EDA) involved examining the distribution of the target variable. The results revealed a notable class imbalance with approximately 80.9% of observations classified as non-default cases and 19.1% classified as default cases.



Such an imbalance is commonly observed in credit risk datasets because the majority of customers fulfil their repayment commitments. Nevertheless, imbalanced class distributions can affect model learning and may result in biased predictions favouring the majority class.

1.3 Statistical Summary

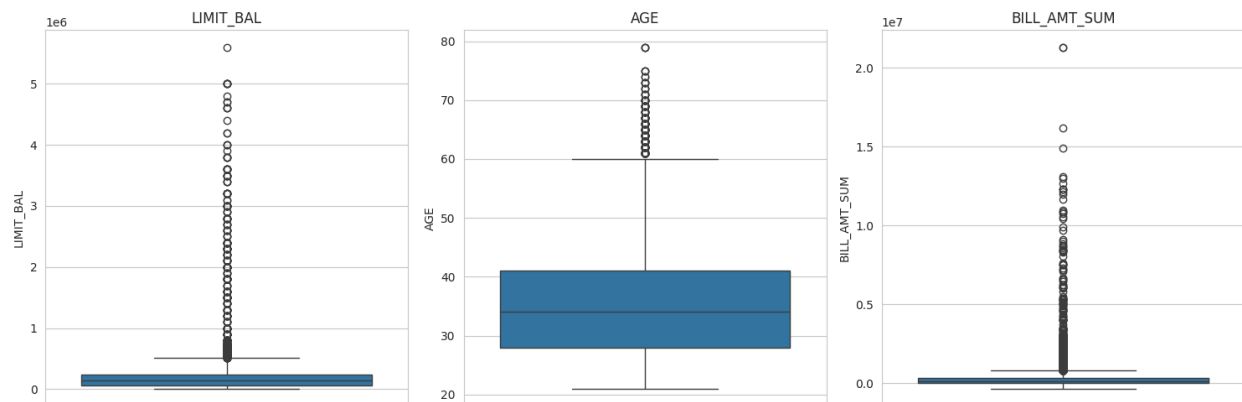
Descriptive statistical analysis was performed on several key numerical features including **LIMIT_BAL**, **AGE**, **BILL_AMT_SUM**, **LIMIT_BAL_LOG**, and **risk_leak**.



The findings highlighted considerable variability in customer credit limits and billing amounts. Furthermore, several variables displayed right-skewed distributions, indicating that a relatively small group of customers held substantially higher balances compared with the wider customer population.

1.4 Identification of Outliers

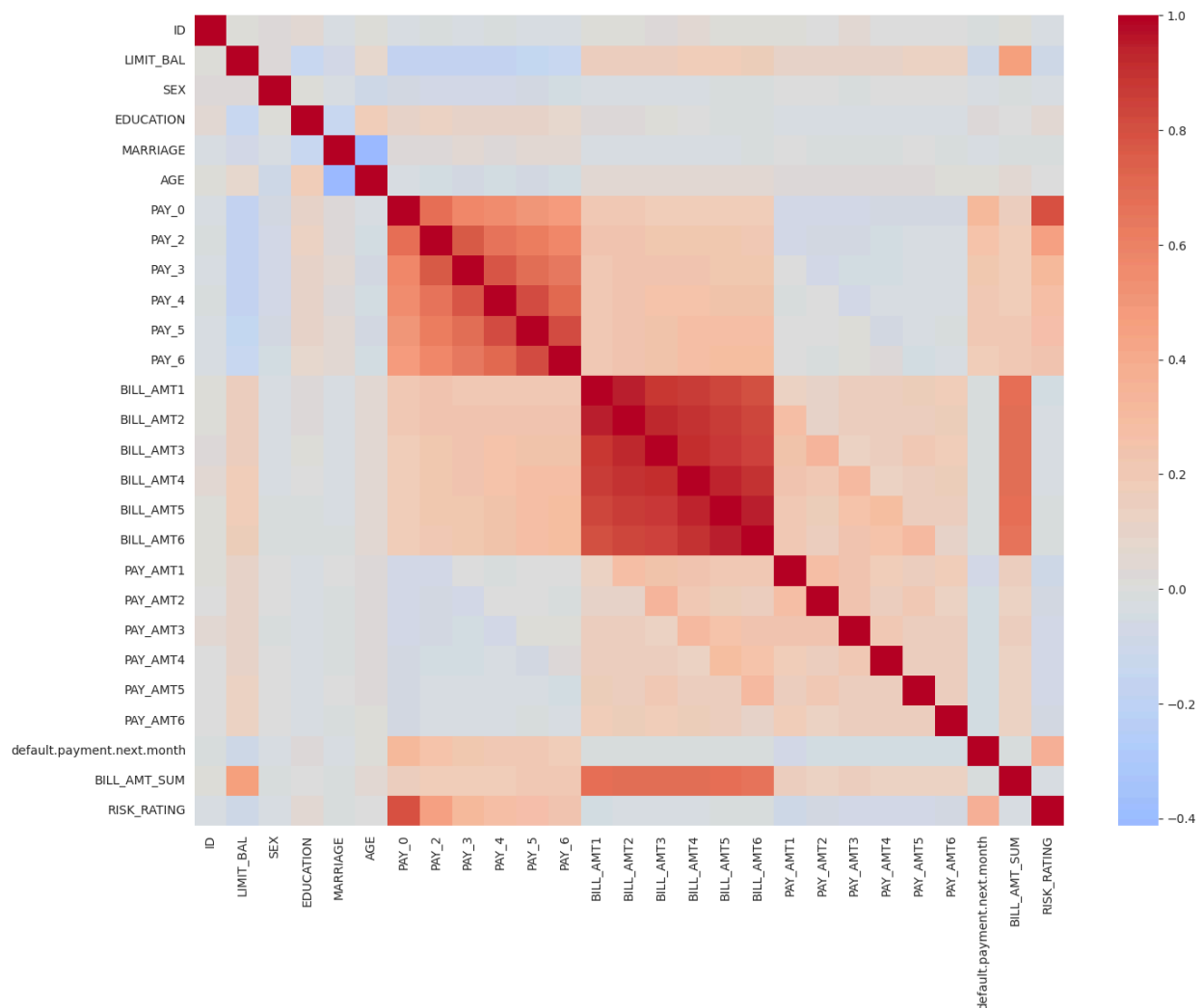
Outlier detection was performed through boxplot analysis and the Interquartile Range (IQR) method to identify observations that deviated significantly from the typical distribution of the data.



Several extreme values were detected in variables including **LIMIT_BAL** and **BILL_AMT_SUM**. Given the nature of financial datasets, such values are expected and may represent customers with unusually high credit exposure or spending patterns. As these observations are likely to be valid and informative, they were retained to preserve the integrity and representativeness of the dataset.

1.5 Correlation Assessment

A correlation matrix was generated to evaluate the relationships between variables and identify potential patterns within the dataset.



The results demonstrated that several repayment status variables appeared moderately to strongly correlated, suggesting that historical payment behaviour may be associated with future default outcomes. This observation aligns with existing credit risk research which consistently identifies behavioural indicators as moderate predictors of default than demographic characteristics.

1.6 Data Preprocessing

Data preparation was undertaken to ensure the dataset was suitable for machine learning modelling. Missing values were addressed using median imputation as this method is more robust to skewed distributions and outliers than mean-based approaches. To improve model efficiency and maintain consistency across variables, numerical features were standardised using the StandardScaler technique.

The **ID** attribute was removed because it served solely as a unique identifier and provided no predictive value. Retaining such variables can introduce unnecessary noise and negatively affect model performance.

Finally, the dataset was partitioned into training and testing subsets using an 80:20 split. Stratified sampling was employed to preserve the original class proportions in both datasets, thereby ensuring a representative evaluation of model performance.

CHAPTER TWO

Machine Learning Models and Evaluation

2.1 Selection of Machine Learning Models

Five classification algorithms were chosen to predict credit card default risk:

1. Logistic Regression
2. Support Vector Machine (SVM)
3. Decision Tree
4. Random Forest
5. K-Nearest Neighbours (KNN)

These algorithms were selected because they represent diverse machine learning paradigms including linear, nonlinear, tree-based and instance-based approaches. Comparing multiple models enables a comprehensive assessment of their effectiveness and helps identify the most appropriate technique for credit risk prediction.

2.2 Model Performance Assessment

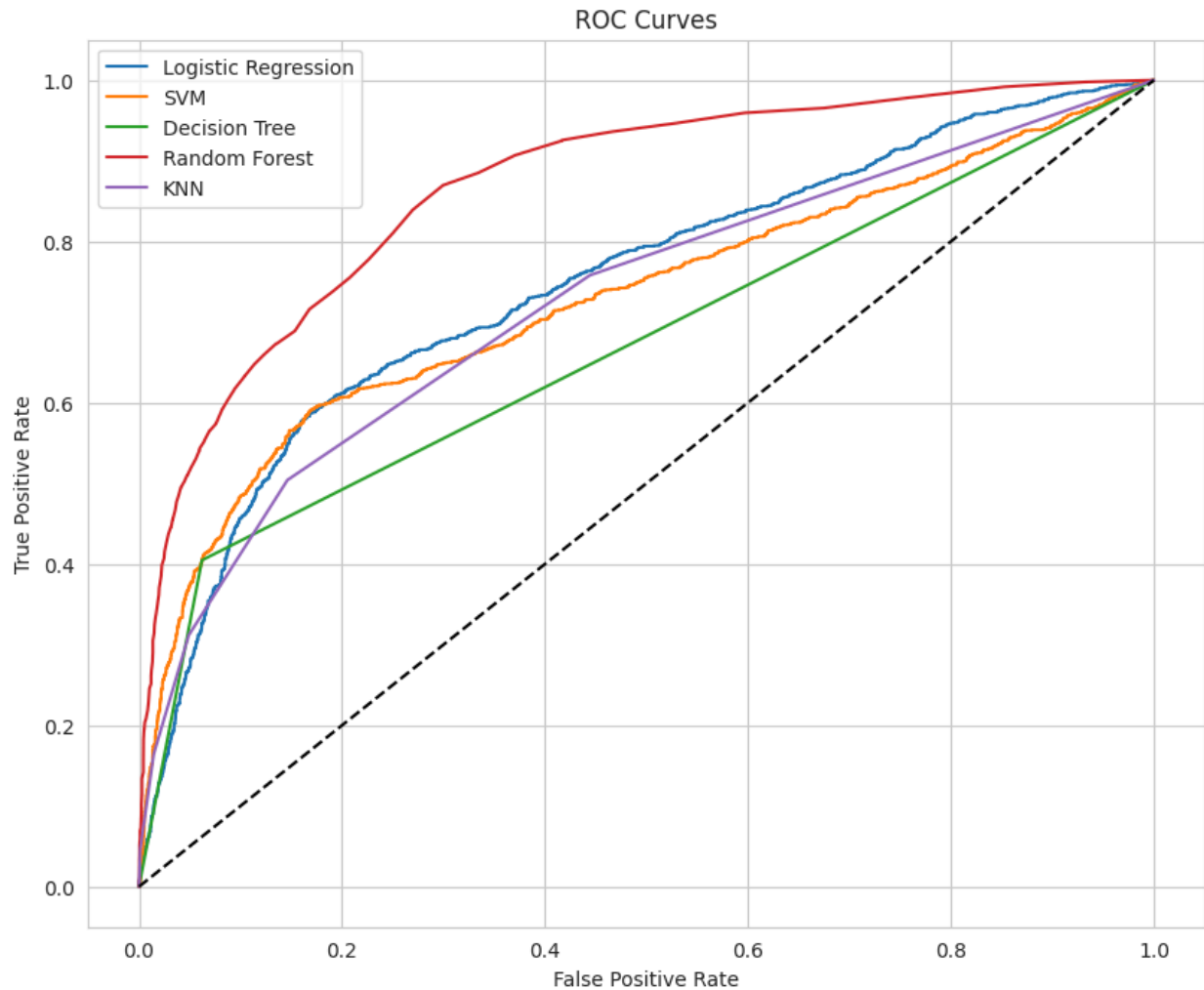
The predictive performance of each model was evaluated using several classification metrics including Accuracy, Precision, Recall, F1-Score and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

Table : Classification Performance Results

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)	ROC-AUC
Logistic Regression	82.16	58.86	22.46	32.52	0.747
Support Vector Machine (SVM)	83.90	66.61	31.78	43.03	0.728
Decision Tree	83.48	60.20	40.34	48.31	0.670
Random Forest	85.81	81.73	33.28	47.30	0.869
K-Nearest Neighbours (KNN)	82.87	60.06	31.18	41.05	0.726

The Random Forest classifier outperformed all other models across most evaluation metrics. It achieved the highest levels of accuracy, precision, F1-score and ROC-AUC demonstrating superior predictive capability. This strong performance can be attributed to the ensemble nature of Random Forest which combines multiple decision trees through bootstrap aggregation to reduce variance and enhance generalisation performance (Breiman, 2001).

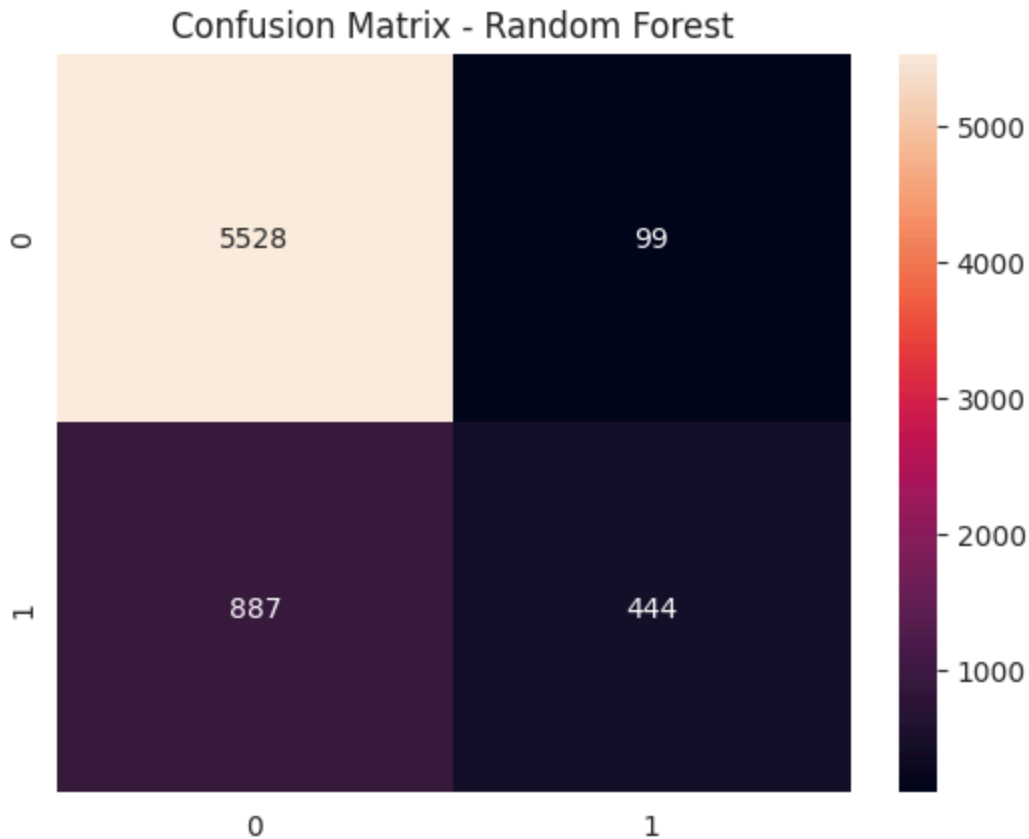
2.3 Receiver Operating Characteristic (ROC) Analysis



ROC curve analysis was conducted to evaluate each model's ability to distinguish between default and non-default customers. The curves illustrate the relationship between the true positive rate and false positive rate across different classification thresholds.

Among the evaluated models, Random Forest achieved the largest ROC-AUC value indicating the strongest discriminatory power and the most effective separation of high-risk and low-risk customers.

2.4 Confusion Matrix Evaluation



Confusion matrix analysis was used to examine classification outcomes in greater detail. While the Random Forest model successfully classified the majority of observations, a proportion of default cases remained incorrectly identified as non-defaults. This limitation is reflected in the comparatively moderate recall scores observed across all classification models.

2.5 Hyperparameter Tuning

To enhance predictive performance, Random Forest hyperparameters were optimised using GridSearchCV with five-fold cross-validation.

The optimal parameter combination identified was:

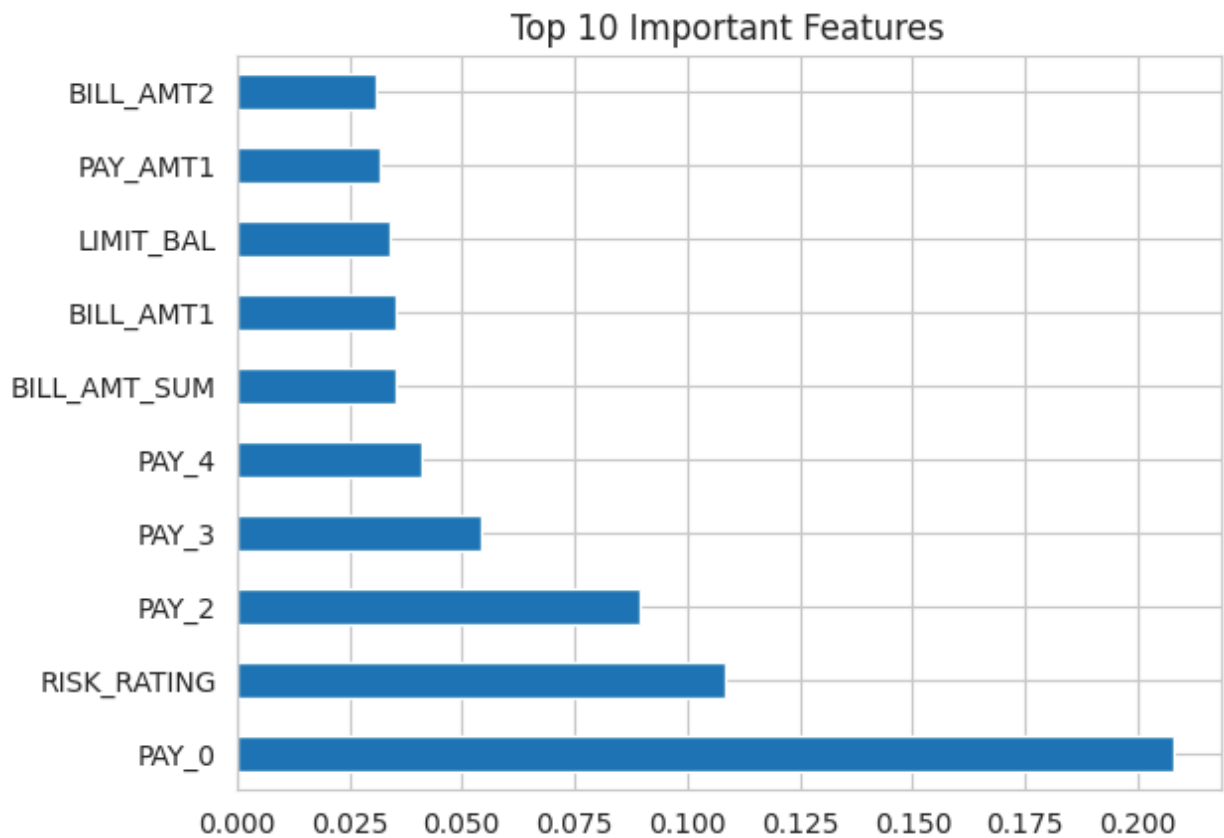
Maximum tree depth (**max_depth**) = 10

Number of estimators (**n_estimators**) = 100

The tuned model achieved a mean cross-validation ROC-AUC score of 0.778. Limiting tree depth helped control model complexity, thereby reducing overfitting while maintaining strong predictive accuracy.

2.6 Feature Importance and Explainability

Feature importance analysis was performed using the best-performing Random Forest model to identify the variables that contributed most significantly to prediction outcomes.

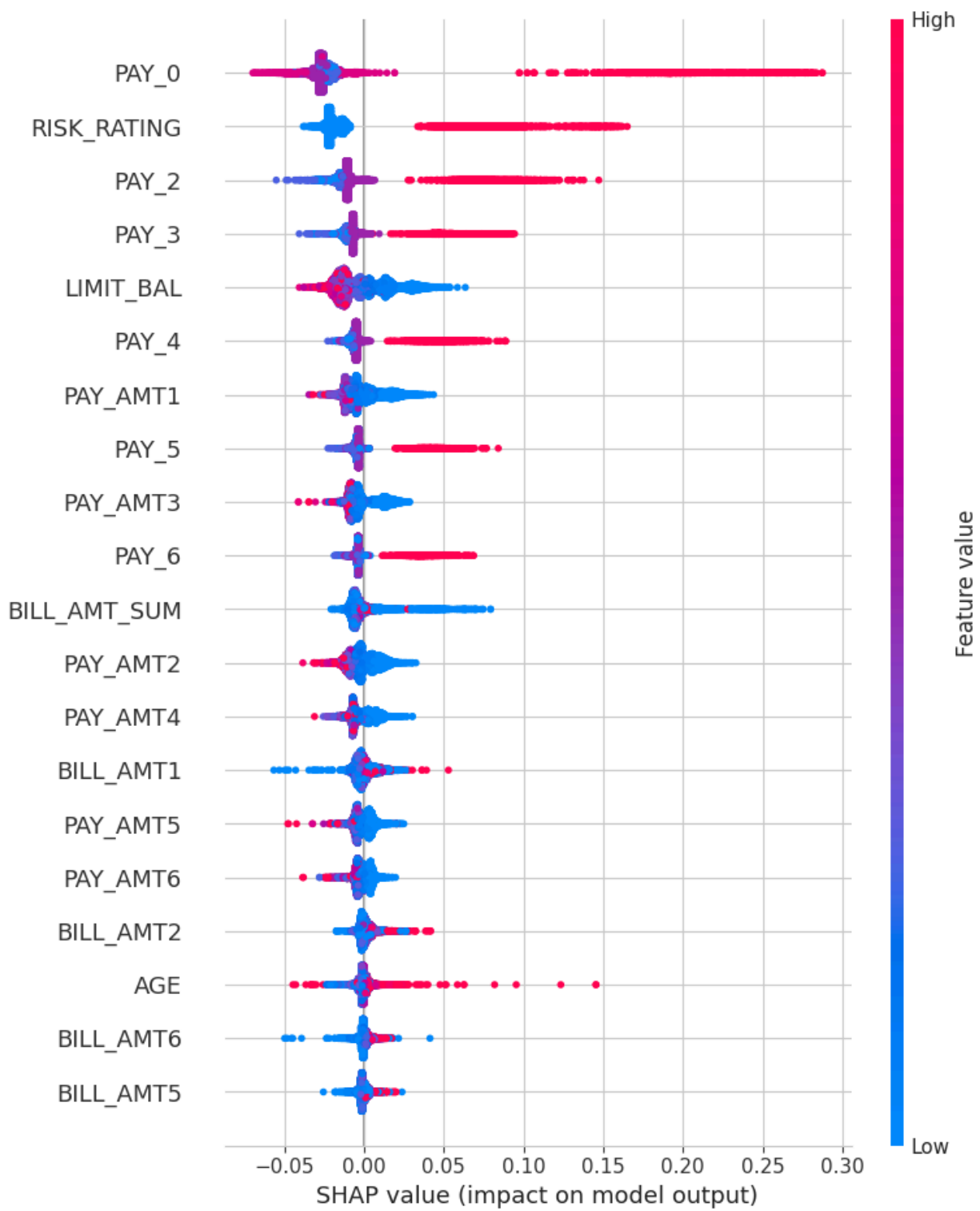


The ten most influential predictors were:

1. PAY_0
2. RISK_RATING
3. PAY_2
4. PAY_3
5. BILL_AMT_SUM
6. BILL_AMT1
7. PAY_4
8. PAY_5
9. LIMIT_BAL
10. BILL_AMT2

Among these variables, **PAY_0** emerged as the most important predictor suggesting that recent repayment behaviour is highly indicative of future default risk. In general, repayment history features and engineered risk indicators contributed more strongly to model performance than demographic characteristics.

To improve model transparency and interpretability, SHAP can be employed to the optimised Random Forest model in order to quantify and visualise the contribution of individual features to specific predictions (Lundberg & Lee, 2017).



2.7 Business Implications

The findings of this study demonstrate the practical usefulness of machine learning for credit risk evaluation in financial institutions. The Random Forest model showed strong predictive performance suggesting that such approaches can effectively help banks identify customers who are more likely to default.

In real-world applications, these results can support improved lending decisions through more targeted risk management actions including credit limit adjustments, closer monitoring of high-risk clients and faster approval processes for low-risk customers.

The feature importance analysis indicates that repayment behaviour is the most significant driver of default prediction. This implies that financial institutions should place greater emphasis on behavioural data when evaluating creditworthiness.

Overall, machine learning-based credit scoring models can enhance decision-making processes, reduce financial exposure and enable more efficient and data-driven lending strategies.

2.8 Study Limitations

Several limitations should be considered when interpreting the findings of this study.

Firstly, the imbalanced nature of the dataset may have influenced model performance particularly with respect to recall and in turn, making it more challenging to correctly identify all default cases. Moreover, although median imputation is a robust approach for handling missing values, it may not fully capture the underlying characteristics of missing observations. Furthermore, the inclusion of demographic variables raises potential fairness and ethical concerns as such attributes may inadvertently contribute to biased lending decisions if not carefully monitored.

Finally, the dataset reflects historical customer behaviour captured within a specific context. Consequently, model performance may vary when applied to different customer populations, geographic regions or economic conditions limiting the applicability of the results.

CONCLUDING REMARKS



This project explored the use of machine learning methods to predict credit card default risk. By conducting exploratory data analysis, data preprocessing, model development and performance evaluation, the study successfully identified customer repayment behaviour as the most significant factor influencing the likelihood of future default.

Among the five classification models assessed, the Random Forest algorithm delivered the strongest overall results achieving an accuracy of 85.81% and a ROC-AUC score of 0.869. The feature importance analysis further revealed that recent repayment status, historical payment patterns and engineered risk-related variables were the key contributors to accurate default prediction.

Future research could focus on improving model performance by addressing class imbalance through techniques such as SMOTE and class-weighted learning. In addition, fairness and bias assessments should be incorporated to support the responsible and ethical use of predictive models in financial decision-making. Further experimentation with advanced ensemble learning methods including XGBoost, LightGBM and CatBoost may also enhance the predictive accuracy and provide valuable insights for future credit risk modelling applications.

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Colab File Link ->

<https://colab.research.google.com/drive/1BhUKZS-LedCLPnf0ZdASrnD7eNG2GFP6#scrollTo=Ss14cVyVi1OD>

Github link ->

<https://github.com/MuhtasimFuad069/Introduction-to-Artificial-Intelligence>